USC Dornsife
Dana and David Dornsife
College of Letters, Arts and Sciences
Spatial Sciences Institute

SSCI 135 (35619-), ~~-~~Maps in the Digital World

Syllabus

Units: 4

Term Day Time: Spring 2026
Lecture: Mon Wed Fri 10:00-10:50am PT
Discussions: Tue 11:00-11:50am PT
Thu 1:00-1:50pm PT~~11:00-11:50am PT~~

Location: GFS 101 (lecture)

Instructor: Diana Ter-Ghazaryan, PhD, GISP
Office: AHF B55K
Regular Office Hours: Monday, 11:00am-1:00pm PT
Also available most days and times by appointment via email.
Contact Info: terghaza@usc.edu, 213-821-1190

Teaching Assistant: TBD
Office: AHF B56A
Office Hours: TBD
Contact Info: TBD

Library Help: Andy Rutkowski
Office: LIPA B40-A
Office Hours: TBD
Contact Info: arutkows@usc.edu

IT Help: Spatial Support
Contact Info: spatial_support@usc.edu.

Course Scope and Purpose

This course explores all the ways in which maps are being used to compile, build, and share knowledge of the world around us. The first maps appeared long ago and today maps are used extensively across the physical, life, and social sciences as well as the humanities. Numbers and quantitative data feature prominently in the preparation of most maps. The overarching intent of this course is to examine some of the ways in which formal reasoning, abstract representation, and empirical analysis are used to construct the maps that you see and use in a given field of study and in everyday life. The topics covered in this course will range from geodetic principles (the way location is measured on the Earth's surface) to the various ways in which information is captured and represented on maps, the role of scale and map projections, and the ways in which various hierarchies and classifications can be combined and used with empirical analysis to add meaning to maps. This course is a Quantitative Reasoning General Education course. Maps are used to engage you in the analysis and manipulation of data and information related to quantifiable objects, symbolic elements, and logic to help navigate the complexity and sophistication of the contemporary world. The lectures and accompanying homework assignments will focus on the role of maps in contemporary life and how numbers are used to construct maps of the world around us. The assignments and final project will increase your capacity to evaluate chains of formal reasoning (the use of formal logic and mathematics), abstract representation (the use of symbolic and diagrammatic representations), and empirical analysis (the use of statistical inference) in building and interpreting various kinds of maps.

Learning Outcomes

On completion of this course, students should be able to:

- Describe and interpret the complexity and sophistication of maps and mapping in the modern world.
- Use a set of formal tools, including logical and statistical inference, probability and mathematical analysis, to pose and evaluate hypotheses, claims, questions, or problems with a variety of maps.
- Distinguish between the assumptions and implications for the logical structures embedded in various kinds of maps.
- Identify both useful and specific applications of various kinds of maps.

Prerequisite(s): None
Co-Requisite(s): None

Class Conduct

Harassment, sexual misconduct, interpersonal violence, and stalking are not tolerated by the university. All faculty and most staff are considered Responsible Employees by the university and must forward all information they receive about these types of situations to the ~~Title IX~~Civil

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Rights Coordinator. The Civil Rights~~Title IX~~ Coordinator is responsible for assisting students with supportive accommodations, including academic accommodations, as well as investigating these incidents if the reporting student wants an investigation. The ~~Title IX~~Office of Civil Rights Compliance (OCRC) ~~office~~ is also responsible for coordinating supportive measures for transgender and nonbinary students such as faculty notifications, and more. If you need supportive accommodations, you may contact the ~~Title IX Coordinator~~ OCRC directly (ocrc@usc.edu~~titleix@usc.edu~~ or 213-~~821-8298~~740-5086) without sharing any personal information with me. If you would like to speak with a confidential counselor, Relationship and Sexual Violence Prevention Services (RSVP) provides 24/7 confidential support for students (213-740-9355 (WELL); press 0 after hours.

USC Culture Journey is an initiative aimed at fostering a more **inclusive, values-based, and integrity-driven culture** across the university. The components of the Culture Journey include: community engagement, core values, strategic cultural goals, ongoing assessment, and leadership commitment. Use the following link to see full details on the Culture Journey: https://culturejourney.usc.edu/

~~**Diversity and Inclusion –** It is my intent that students from all diverse backgrounds and perspectives be well served by this course, that students' learning needs be addressed both in- and out of class, and that the diversity that students bring to this class be viewed as a resource, strength and benefit. It is my intent to present materials and activities that are respectful to everyone, and you are also expected to respect of others regardless of their race, ethnicity, gender identity and expressions, cultural beliefs, religion, sexual orientation, national origin, age, abilities, ideas and perspectives, or socioeconomic status. Your suggestions are encouraged and appreciated. Feel free to let me know ways to improve the effectiveness of the course for you personally or for other students.~~

Course Content Distribution and Synchronous Session Recordings Policies

USC has policies that prohibit recording and distribution of any synchronous and asynchronous course content outside of the learning environment.

Recording a university class without the express permission of the instructor and announcement to the class, or unless conducted pursuant to an Office of Accessibility Services (OSAS) accommodation. Recording can inhibit free discussion in the future, and thus infringe on the academic freedom of other students as well as the instructor. (Living our Unifying Values: The USC Student Handbook, page 13).

Distribution or use of notes, recordings, exams, or other intellectual property, based on university classes or lectures without the express permission of the instructor for purposed other than individual or group study is prohibited. This includes but is not limited to providing materials for distribution by services publishing course materials. This restriction on unauthorized use also applies to all information, which has been distributed to students or in any way has been displayed for use in relationship to the class, whether obtained in class, via email, on the internet, or via any other media. (Living our Unifying Values: The USC Student Handbook, page 13).

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Technological and Communication Requirements

ArcGIS Online is a web-based analysis platform and access to it will be provided at the start of the semester. Students do not need to install anything on their own computer to access ArcGIS Online. Instead, every student must have the following technology requirements:

- A computer with a fast internet connection
- An up–to-date-web browser to access ArcGIS Online (Chrome usually works best).

If a student does not have access to any of these, please speak with the instructor at the start of the semester. And see the USC ITS Student Toolkit here: https://keepteaching.usc.edu/students/student-toolkit/

SSI Tech Support – This course utilizes ArcGIS Online, which is a web-based mapping platform. If students are unable to log into ArcGIS Online or experience technical issues, they should send an email (via their USC account) to SSI Tech Support at spatial_support@usc.edu, making sure to copy (cc) the instructor on the email. Assignment specific questions should be directed to the instructor.

Communications – All materials to be handed in will be submitted via Brightspace. It is each student's responsibility to stay informed about what is going on in our course. In addition to email about time-sensitive topics, any important announcements will be posted on the Announcement page in Brightspace.

Be sure to check these each time you log onto Brightspace. I will send via email through Brightspace any notices that are time sensitive. Please be sure that you read as soon as possible all email sent from Brightspace or from me. Do not ignore course email until the day before assignments are due. Also, double check to be sure that email sent from the USC blackboard account does not go into your junk mail!

While I am usually on-line all day and will probably respond to emails from students very quickly, I will endeavor to respond to all email within 24 hours of receipt, aiming for no more than 72 hours delay. In the rare case when I expect to be off-line for more than 72 hours, I will post an announcement on the Brightspace site.

Required Readings and Supplementary Materials

The required textbook for this course is:

- Arlinghaus, S. L. and J. Kerski. 2014. *Spatial Mathematics: Theory and Practice through Mapping*. Boca Raton, FL: CRC Press.

Description of Assessments and Assignments

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Your grade in this class will be determined on the basis of several different assessments:

Homework Assignments – 10 worth a total of 50 points. In addition to lectures and in-class discussions, there is a series of homework assignments that are designed to introduce the tools of quantitative reasoning and provide practical experience in implementing these tools to explore various problems within the framework of the scientific method. These assignments are linked to the lectures and class discussions, but do not duplicate the classroom experience. Home assignments will be graded and returned, and the midterm and final exams will have a home assignment component to them. In other words, the home assignments are an important and integral part of the course as a whole. Homework assignments include:

- Mental Maps
- Wild Maps
- Routes and Coordinates
- Distance
- Projections
- Symbols
- Build a Map
- GNSS
- Remote Sensing
- Central Tendency

In-Class Activities – 13 worth 2 points. This class contains many in-class activities that take place during the discussion sessions. You will receive 1 point for completion, and 1 point for accuracy/effort. You cannot make-up these assignments if you are not in class and do not notify me or our TA of your absence in advance.

Midterm Examination – 1 worth 12 points. The mid-term examination will consist of multiple-choice, short answer, and essay questions. Students will be expected to take the exam at the indicated time.

Final Examination – 1 worth 12 points. The final examination will consist of multiple-choice, short answer, and essay questions. Students will be expected to take the exam at the indicated time.

Assignment Submission and other Policies

Assignments must be submitted via Blackboard by the due dates specified in the Course Schedule. Attention to on-time assignment submission is essential. The instructor will aim to return feedback before the next assignment is due. Strict penalties apply for late assignments as follows:

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- Assignments up to four-days late will receive a 1-point deduction per day. No assignments submitted after four days will be accepted or graded.
- Additionally, no written work will be accepted for grading after 5 pm PT on the last day of classes.
- Absences from class sessions must be requested by sending an email to the instructor. Excused absences from class sessions will be granted only for valid reasons; please notify me of the reason for your absence in your email.

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Grading Breakdown

Assessment	Number	Points Each	Total Points
Home Assignments (HA)	10	5	50
Pop attendance in lecture	3	2	6
Discussion	12~~3~~	2	24~~6~~
Midterm Examination	1	10~~2~~	10~~2~~
Final Exam	1	10~~2~~	10~~2~~
Total	24	-	100

Grading Scale

Assignments in this and other SSCI courses, are graded on the letter grade scale where A is exemplary, B is very good, C is satisfactory, D is unsatisfactory, and F needs improvement. The grading scale follows:

A	> 93 points		B-	80-82 points		D+	67-69 points
A-	90-92 points		C+	77-79 points		D	63-66 points
B+	87-89 points		C	73-76 points		D-	60-62 points
B	83-86 points		C-	70-72 points		F	<60 points

Schedule

Week	Topic	Readings	Deliverables/Due Dates
Module 1: Introducing Spatial Sciences			
Week 1 (1/12-1/16)	Monday: Welcome to Spatial Sciences!		
	Wednesday: Why Maps?		
	Friday: Mental Maps		
Module 2: Measuring Earth			
Week 2 (1/19-1/23)	* Monday: MLK Day-no class	Arlinghaus and Kerski, Ch. 1	
	Wednesday: Round Earth		HA #1 Due 1/23
	Friday: Location and Direction		
Week 3 (1/26-1/30)	Monday: Latitude and Longitude	Arlinghaus and Kerski, Ch. 2	
	Wednesday: Seasons		HA #2 Due 1/30
	Friday: Antipodes and Degrees		
Week 4 (2/2-2/6)	Monday: Creating a Model of the Earth		
	Wednesday: Other Coordinate Systems		HA #3 Due 2/6

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Week	Topic	Readings	Deliverables/Due Dates
	Module 3: From the Globe to the Map		
	Friday: Introducing Map Projections		
Week 5 (2/09-2/13)	Monday: Projections Based on Distortion	Arlinghaus and Kerski, Ch. 9	
	Wednesday: Projections Based on Developable Surface		HA #4 Due 2/13
	Friday: Choosing Projections		
Week 6 (2/16-2/20)	*Monday: University Holiday, No Class	Arlinghaus and Kerski, Ch. 5	
	Wednesday: Scale		HA #5 Due 2/20
	Friday: Reviewing Algebra and Fractions		
Week 7 (2/23-2/27)	Monday: Representative Fractions		
	Wednesday: Midterm Review		
	Friday: Midterm Exam		Midterm @ 10:00
Module 4: Making Maps			
Week 8 (3/2-3/6)	Monday: Introducing GIS	Arlinghaus and Kerski, Ch. 3 Arlinghaus and Kerski, Ch. 4	
	Wednesday: Data Models/Rasters and Vectors		
	Friday: GIS Applications		
Week 9 (3/09-3/13)	Monday: Intro to Cartography	Arlinghaus and Kerski, Ch. 6	
	Wednesday: Types of Maps		HA #6 Due 3/13
	Friday: Types of Maps Continued		
Spring Recess (3/16-3/20)	*No Class		
Week 10 (3/23-3/27)	Monday: Choropleth Maps		
	Wednesday: Dot Density Maps, Isopleth Maps and others		HA #7 Due 3/27
	Friday: Color		
Module 5: From a Distance-GNSS and Remote Sensing			
Week 11 (3/30-4/3)	Monday: GNSS	"What is Remote Sensing?" NASA	
	Wednesday: GNSS Continued		HA #8 Due 4/3
	Friday: Remote Sensing		
Week 12 (4/6-4/10)	Monday: Remote Sensing and Calculating Speed		
	Module 6: Spatial Patterns		

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Week	Topic	Readings	Deliverables/Due Dates
	Wednesday: Introduction to Spatial Analysis	Arlinghaus and Kerski, Ch. 8	
	Friday: Intro to Spatial Analysis Contd		
Week 13 (4/13-4/17)	Monday: Data Types and Central Tendency		
	Wednesday: Clustering		
	Friday: Clustering Contd		HA #9 Due 4/17
Week 14 (4/20-4/24)	Monday: Density Analysis and Quadrats		
	Wednesday: Geoenrichment and Overlay		
	Friday: Site Selection		
Week 15 (4/27-5/1)	Monday: Distance Statistics		
	Wednesday: Distance Statistics Contd		
	Friday: Wrap Up and Final Review		HA #10 Due 5/1
	Final Exam – TBD		

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Statement on Academic Conduct and Support Systems

Academic Integrity

The University of Southern California is a learning community committed to developing successful scholars and researchers dedicated to the pursuit of knowledge and the dissemination of ideas. Academic misconduct, which includes any act of dishonesty in the production or submission of academic work, comprises the integrity of the person who commits the act and can impugn the perceived integrity of the entire university community. It stands in opposition to the university's mission to research, educate, and contribute productively to our community and the world.

All students are expected to submit assignments that represent their own original work, and that have been prepared specifically for the course or section for which they have been submitted. You may not submit work written by others (including AI generated) or "recycle" work prepared for other courses without obtaining written permission from the instructor(s).

Other violations of academic integrity include, but are not limited to, cheating, plagiarism, fabrication (e.g., falsifying data), collusion, knowingly assisting others in acts of academic dishonesty, and any act that gains or is intended to gain an unfair academic advantage.

The impact of academic dishonesty is far-reaching and is considered a serious offense against the university. All incidences of academic misconduct will be reported to the Office of Academic Integrity and could result in outcomes such as failure on the assignment, failure in the course, suspension, or even expulsion from the university.

For more information about academic integrity see the student handbook or the Office of Academic Integrity's website, and university policies on Research and Scholarship Misconduct.

Please ask your instructor if you are unsure what constitutes unauthorized assistance on an exam or assignment, or what information requires citation and/or attribution.

SSI Policy on the Creation of Original Work and Use of Generative AI

All students are expected to submit assignments that represent their own original work, and that have been prepared specifically for the course or section for which they have been submitted. Students may not have another person or entity complete any substantive portion of an assignment or reuse work prepare for courses without obtaining written permission from the instructor(s). Developing strong competencies in research, writing, and the technical execution of geospatial technologies are foundational to SSI academic programs that are designed to prepare you for success in the workplace. Therefore, using generative AI tools – unless explicitly specified otherwise – is strictly prohibited in this course, will be identified as plagiarism, and will be reported to the Office of Academic Integrity.

Students and Disability Accommodations:

USC welcomes students with disabilities into all of the University's educational programs. The Office of Student Accessibility Services (OSAS) is responsible for the determination of appropriate accommodations for students who encounter disability-related barriers. Once a student has completed the OSAS process (registration, initial appointment, and submitted

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documentation) and accommodations are determined to be reasonable and appropriate, a Letter of Accommodation (LOA) will be available to generate for each course. The LOA must be given to each course instructor by the student and followed up with a discussion. This should be done as early in the semester as possible as accommodations are not retroactive. More information can be found at osas.usc.edu. You may contact OSAS at (213) 740-0776 or via email at osasfrontdesk@usc.edu.

~~**Support Systems:**~~

Support Systems:

Counseling and Mental Health - (213) 740-9355 – 24/7 on call

Free and confidential mental health treatment for students, including short-term psychotherapy, group counseling, stress fitness workshops, and crisis intervention.

988 Suicide and Crisis Lifeline - 988 for both calls and text messages – 24/7 on call

The 988 Suicide and Crisis Lifeline (formerly known as the National Suicide Prevention Lifeline) provides free and confidential emotional support to people in suicidal crisis or emotional distress 24 hours a day, 7 days a week, across the United States. The Lifeline is comprised of a national network of over 200 local crisis centers, combining custom local care and resources with national standards and best practices. The new, shorter phone number makes it easier for people to remember and access mental health crisis services (though the previous 1 (800) 273-8255 number will continue to function indefinitely) and represents a continued commitment to those in crisis.

CARE-SC - (213) 740-9355(WELL) – 24/7 on call

Free and confidential therapy services, workshops, and training for situations related to gender- and power-based harm (including sexual assault, intimate partner violence, and stalking).

Office for Civil Rights Compliance - (213) 740-5086

Information about how to get help or help someone affected by harassment or discrimination, rights of protected classes, reporting options, and additional resources for students, faculty, staff, visitors, and applicants.

The Office of Student Accessibility Services (OSAS) - (213) 740-0776

OSAS ensures equal access for students with disabilities through providing academic accommodations and auxiliary aids in accordance with federal laws and university policy.

USC Campus Support and Intervention - (213) 740-0411

Assists students and families in resolving complex personal, financial, and academic issues adversely affecting their success as a student.

USC Culture Journey - (213) 740-8258

At USC our six unifying values support our University's mission, and guide our behaviors, the way we engage with ourselves and our community, to create our Trojan culture.

USC Emergency - UPC: (213) 740-4321, HSC: (323) 442-1000 – 24/7 on call

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Emergency assistance and avenue to report a crime. Latest updates regarding safety, including ways in which instruction will be continued if an officially declared emergency makes travel to campus infeasible.

USC Department of Public Safety - UPC: (213) 740-6000, HSC: (323) 442-1200 – 24/7 on call

Non-emergency assistance or information.

Office of the Ombuds - (213) 821-9556 (UPC) / (323-442-0382 (HSC)

A safe and confidential place to share your USC-related issues with a University Ombuds who will work with you to explore options or paths to manage your concern.

Occupational Therapy Faculty Practice - (323) 442-2850 or otfp@med.usc.edu

Confidential Lifestyle Redesign services for USC students to support health promoting habits and routines that enhance quality of life and academic performance.

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